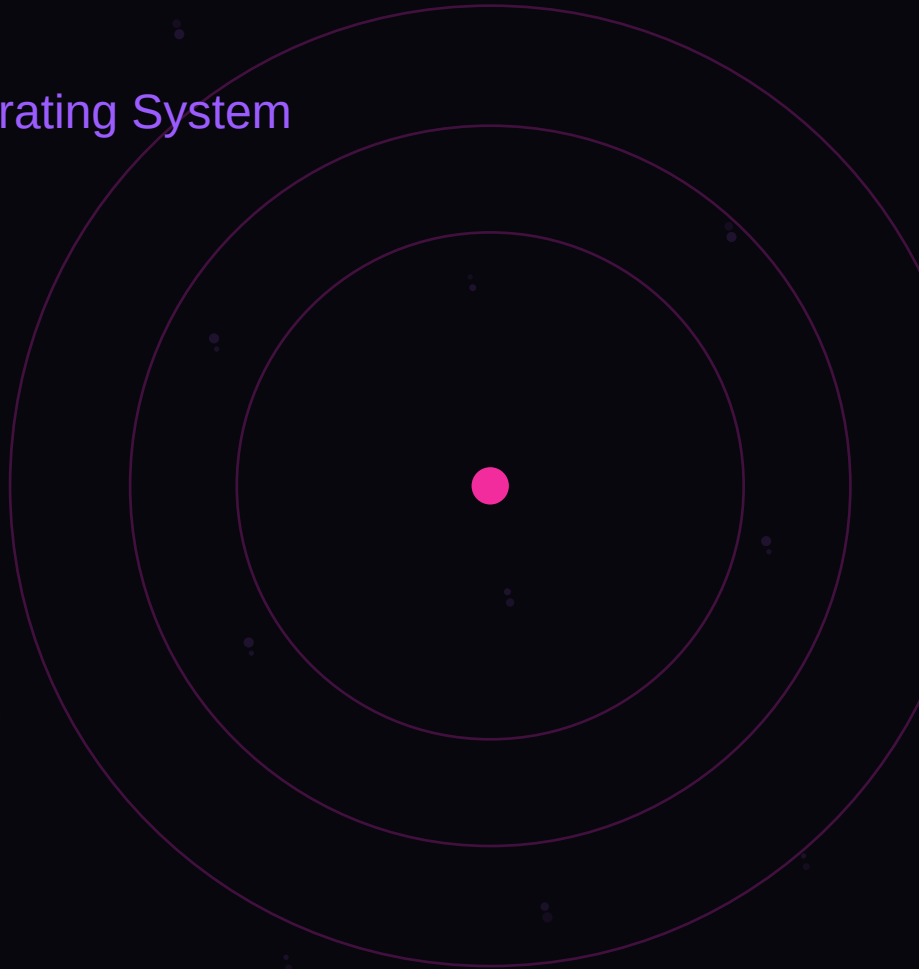


MASTER BLUEPRINT / 2026

LILITH OS

Personal Intelligence Operating System

ARCHITECTURE v0.1



A persistent, privacy-aware intelligence layer that understands your world, coordinates specialized apps and agents, and presents itself through one living visual shell.

DESIGN PRINCIPLE

THE OS IS THE INTERFACE. LILITH IS THE INTELLIGENCE.

01 / NORTH STAR

A personal intelligence layer, not another chatbot

Persistent context. Structured truth. Visible agency. Human warmth without pretending to be human.

UNDERSTANDS

Entities, relationships, history, preferences, goals and ongoing situations.

COORDINATES

Apps, agents, devices, schedules and external services through one event fabric.

ACTS SAFELY

Read freely where allowed; draft, propose, approve and audit before consequential action.

POSITIONING

LILITH survives any single model, runtime, channel or cloud. Hermes is a reasoning runtime today - not the identity, data model or operating system.

PRODUCT PROMISE

One command center that knows what matters now, explains why, and keeps continuity across career, projects, communication, home and life.

Success feels like continuity: LILITH notices the missing meeting link before you have to.

02 / SYSTEM MAP

Layered architecture

Replaceable runtimes below a stable identity and contract.

EXPERIENCE

3D avatar | Chat | Command palette | Dashboard | Mobile

LILITH CORE

Reasoning | Personality | Context | Planning | Approval

EVENT + AUTOMATION

Events | Watchers | Schedules | Workflows | Attention

INTELLIGENCE DATA

Entity graph | Memory | App state | Truth | Audit history

APPS

Career | Mail | Calendar | Projects | Knowledge | Personal CRM

INTEGRATIONS

Gmail | Calendar | GitHub | Messages | Cloud | Local devices

Contract boundary: UI -> LILITH API -> reasoning/services -> state/event bus -> integrations

03 / INTELLIGENCE

LILITH Core + truth model

The core decides what deserves attention, what is known, and what may happen next.

CONTEXT MANAGER

Assembles the smallest relevant working set.

REASONING ROUTER

Chooses model, tool or specialist agent.

PLANNER

Turns intent into checkpoints and reversible steps.

PERSONALITY LAYER

Shapes tone and presentation, never factual truth.

APPROVAL ENGINE

Gates actions by risk, scope and user policy.

AUDIT EXPLAINER

Records what happened, why and from which evidence.

TRUTH LADDER**OBSERVED**

Raw source evidence

INFERRED

Model-derived; confidence attached

USER-CONFIRMED

Human correction overrides inference

STALE / CONFLICTED

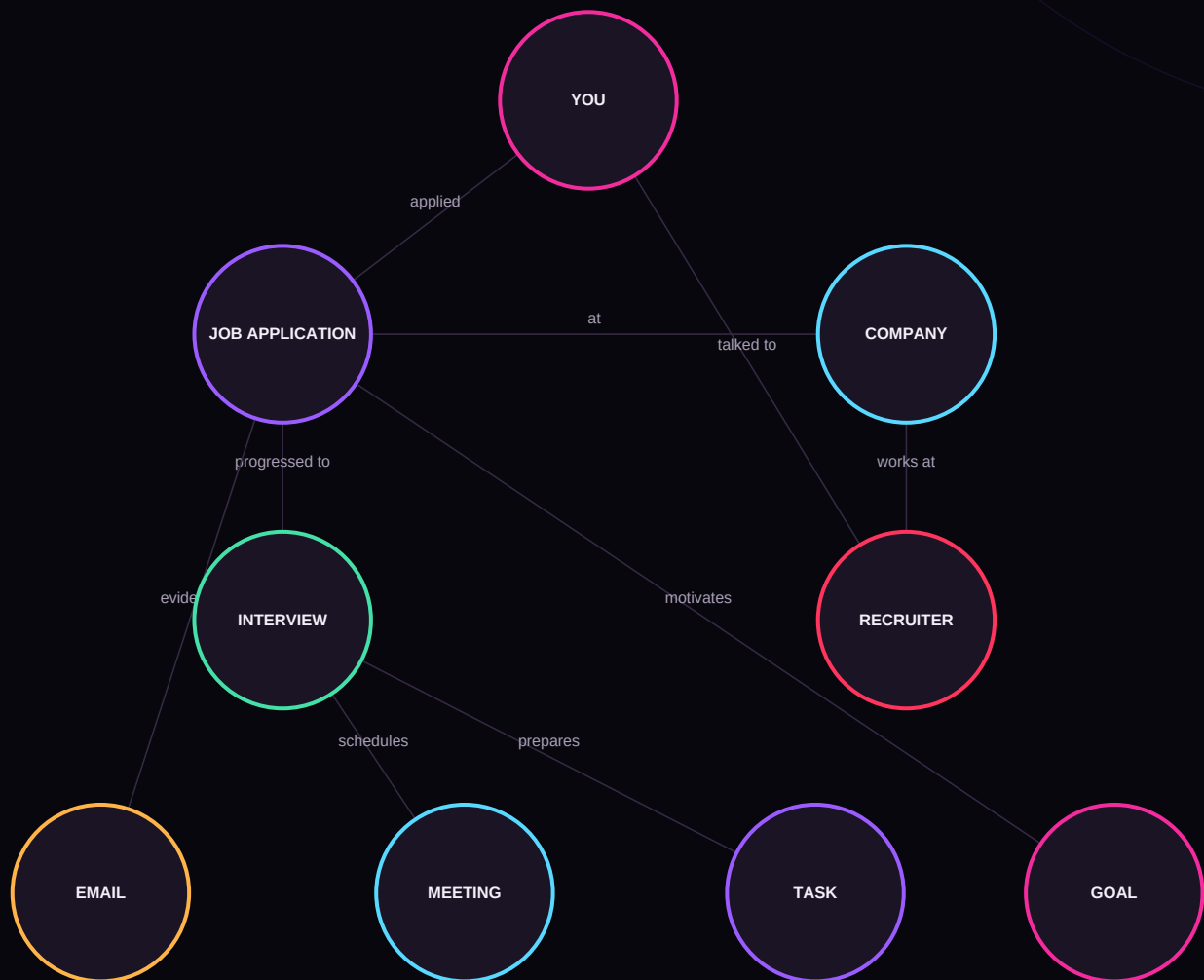
Visible; never silently merged

Rule: personality may color language; it may never color evidence, confidence or permissions.

04 / WORLD MODEL

Entity graph

A connected model replaces isolated JSON fragments without forcing a redesign for every new app.



Extensible types: Person, Company, Role, Conversation, Project, Repository, Document, Device, Automation, Agent - later Expense, Workout, Course, Trip, Server and CloudResource.

05 / NERVOUS SYSTEM

Event bus + attention engine

Everything communicates through typed events. Every event receives an attention score.

EXAMPLE EVENT

career.interview_detected / entity: job_482 / source: gmail / confidence: 0.96 / timestamped

CAREER CRM

stage -> interview

CALENDAR

match meeting

FOLLOW-UP

resolve wait

ATTENTION

priority -> high

VISUAL SHELL

animate notice

0 SILENT

Store only

1 BACKGROUND

Update state

2 PASSIVE

Show in dashboard

3 NOTICE

Mention naturally

4 ALERT

Notify now

5 CRITICAL

Interrupt

Attention is policy + context: urgency, confidence, user state, reversibility and time-to-impact.

06 / PRESENCE

Human-like presentation states

Expressive states communicate system activity. They are presentation, not simulated feelings.

IDLE

ambient

LISTENING

voice active

THINKING

reasoning

RESEARCHING

sources

WORKING

agent active

FOCUSED

deep task

ALERT

urgent

CONCERNED

risk detected

AMUSED

playful reply

PLEASED

success

CELEBRATING

milestone

ANNOYED

friction

SLEEP

low activity

State driver: event + agent lifecycle + attention level -> animation, pose, glow, expression and voice style.

07 / INTERFACE

Visual shell + identity

A calm command surface with LILITH permanently present, not a dashboard full of noise.

HOME

TODAY

CAREER

MAIL

CALENDAR

PROJECTS

MEMORY

AGENTS

AUTOMATE

SYSTEM

Good evening.

Everything is mostly under control.

ATTENTION

Interview tomorrow / confirmation received / prep ready

2 ITEMS

LILITH

ONLINE / FOCUSED

CAREER PIPELINE

Applied 12 Recruiter 4 Interview 3 Offer 0

ASK LILITH...

Voice | command | context

Identity: obsidian field / ultraviolet intelligence / magenta presence / crimson urgency / glass cards.

08 / OPERATIONS

Command center + universal timeline

A single place to inspect state, issue intent and understand every action.

COMMAND PALETTE

Natural-language intent, scoped commands, shortcuts, app actions, agent dispatch and preview-before-run.

LIVE ACTIVITY

Agent status, tool use, confidence, blockers, cost, runtime and cancellation controls.

UNIVERSAL TIMELINE



23:31	Gmail watcher 3 messages scanned
23:31	Career engine Recruiter reply detected
23:31	LILITH Application -> Interview
23:32	Calendar engine Interview event matched
23:32	Follow-up tracker Waiting item resolved

Every entry links to evidence, affected entities, resulting state changes and approval history.

09 / CONTROL

Approval model + security kernel

Autonomy expands by policy, never by ambiguity.

AUTO

LOW RISK

Read, classify, summarize, update reversible local state.

PREVIEW

MEDIUM

Draft messages, proposed calendar changes, workflow plans.

APPROVE

CONSEQUENTIAL

Send, purchase, publish, delete, control devices, change access.

CAPABILITY TOKENS

Narrow, expiring permissions per agent/app.

POLICY ENGINE

Who / what / when / data zone / maximum impact.

SECRETS VAULT

Credentials never enter prompts or app state.

AUDIT LEDGER

Immutable intent, evidence, decision and outcome trail.

KILL SWITCH

Immediate stop for agents, automations and device control.

SANDBOXES

Isolated execution with network and filesystem boundaries.

10 / MEMORY

Memory architecture + privacy zones

Remember deliberately. Separate working context from durable identity and sensitive evidence.

WORKING

Current task context; short-lived.

EPISODIC

Events and interactions over time.

SEMANTIC

Stable facts, preferences and relationships.

PROCEDURAL

Approved routines, policies and successful workflows.

APP STATE

Structured domain records owned by apps.

ARCHIVE

Encrypted raw evidence with retention controls.

PRIVACY ZONES**PUBLIC**

Safe for external models

PERSONAL

Default encrypted profile

SENSITIVE

Career, finance, private communications

SEALED

Secrets, credentials, raw auth data

LOCAL-ONLY

Never leaves trusted device

Memory lifecycle: ingest -> redact -> classify -> consent/policy -> store -> retrieve minimally -> expire/delete.

11 / EXTENSIBILITY

Agent architecture + app framework

Apps own domain truth. Agents perform bounded work. Core coordinates both.

ORCHESTRATOR

Decomposes goals, applies policy, selects specialists and consolidates results.

SPECIALIST AGENTS

Career, research, coding, calendar, finance, home and future capabilities.

EXECUTION WORKERS

Deterministic services, scripts and connectors with narrow permissions.

APP CONTRACT**MANIFEST**

identity, routes, permissions

ENTITIES

owned schemas + relationships

EVENTS

publish / subscribe contracts

VIEWS

cards, pages, timeline renderers

ACTIONS

preview, approve, execute, undo

POLICIES

data zones + risk thresholds

First reference app: Career CRM v1. Later apps plug into the same contracts without rewriting the core.

12 / PLATFORM

Backend, frontend + real-time evolution

Build a small coherent core first; split only when scale and ownership demand it.

BACKEND

Now: Python services + typed APIs + SQLite.
Next: Postgres, job queue, event store, vector retrieval.
Later: independently deployable services.

FRONTEND

Next.js visual shell.
Tauri desktop wrapper.
React + design tokens.
React Three Fiber for avatar scenes.

REAL-TIME

WebSocket/SSE activity stream.
Durable event log.
Optimistic UI with reconciliation.
Presence + agent lifecycle channels.

MODEL ROUTER

Today: LILITH -> Hermes -> current frontier model. Future: policy-driven routing across GPT, Claude, Gemini, local models and specialized agents. Identity and memory remain model-neutral.

OBSERVABILITY

Structured logs, traces, cost, latency, failures, retries and user-visible explanations.

DATA EVOLUTION

JSON -> SQLite -> Postgres + graph projections; migrations preserve app contracts.

13 / EMBODIMENT

Local + cloud hybrid / 3D avatar

Keep sensitive control and continuity local; rent cloud intelligence when it adds value.

LOCAL TRUST ZONE

TRUSTED

Secrets vault
Device control
Privacy redaction
Offline cache
Wake word / microphone gate
Local model fallback

CLOUD INTELLIGENCE

POLICY-GATED

Frontier reasoning
Heavy research
Managed integrations
Sync relay
Remote notifications
Optional rendering

STATE DRIVER

events + lifecycle

AVATAR RUNTIME

Three.js / R3F or game engine

ANIMATION GRAPH

idle, think, work, alert

VOICE PIPELINE

STT -> core -> expressive TTS

LIP + GAZE

phonemes, focus target, presence

FALLBACK

2D / low-power / text-only

The avatar visualizes real system state; it never becomes an authority boundary.

14 / HORIZON

Eventual home experience

LILITH becomes a calm ambient layer across rooms, screens and devices - with explicit presence and control.

DESKTOP

Primary command center and deep work.

MOBILE

Alerts, capture, approvals and location-aware context.

ROOM DISPLAY

Ambient glanceable status and household timeline.

VOICE

Local wake gate, room routing and interruption policy.

HOME HUB

Matter/Home Assistant bridge, routines and device graph.

PERSONAL SERVER

Local memory, event bus, media and resilient automations.

NON-NEGOTIABLE HOME RULES

Physical actions require explicit scopes. Cameras and microphones show active state. Every automation has manual override. Safety systems remain independent.

Goal: useful presence everywhere, invisible surveillance nowhere.

15 / DELIVERY

Foundation status

The nervous system is already alive. These items are the completed base, not speculative roadmap.

OK

Dual-account Gmail watcher

OK

Secondary career inbox summary

OK

Meeting context collector

OK

Morning brief pipeline

OK

Follow-up tracking

OK

Discord alerts

OK

Multi-account handling

OK

Shared sensitive-data redaction

OK

Protected ingestion paths validated

OK

Read-only automation posture

CURRENT GATE

READY

Foundation is operational. Next, normalize contracts and state into OS Core v0.1; then ship Career CRM v1 as the first full app.

16 / ROADMAP

Development phases

Each phase has a demonstrable exit condition, not just a feature list.

0

FOUNDATION

DONE

Watchers, summaries, meeting context, briefs, alerts, redaction.

1

OS CORE v0.1

NEXT

Entity IDs, event envelope, attention, approvals, audit, API.

2

CAREER CRM v1

QUEUED

Pipeline, recruiter graph, timeline, next actions, read-only ingest.

3

VISUAL SHELL

PLANNED

Home, Today, timeline, command palette, design system.

4

AGENTS + APPS

PLANNED

SDK, specialist agents, app manifests, policy scopes.

5

AVATAR + VOICE

HORIZON

State-driven 3D presence and voice interaction.

6

HOME + HYBRID

HORIZON

Local hub, devices, resilient offline/cloud operation.

17 / IMMEDIATE BUILD

OS Core v0.1 checklist

The minimum stable substrate required before the first full app.



Define canonical entity IDs + timestamps

P0



Publish versioned event envelope + schema registry

P0



Create local SQLite state and migration strategy

P0



Implement event log with replay/idempotency

P0



Add attention scoring levels 0-5

P0



Add approval classes: auto / preview / approve

P0



Create audit record: intent -> evidence -> action -> result

P0



Expose typed local API to visual shell

P1



Add health, failure and retry telemetry

P1



Freeze Career CRM app contract

P1

Exit: one synthetic email event flows through ingest -> truth -> attention -> approval -> audit -> UI projection.

18 / FIRST APP

Career CRM v1 checklist

Read-only first. Structured, inspectable and ready for the future visual shell.

- ☐ Schema: company, role, stage, recruiter, source, confidence
- ☐ Stages: discovered -> applied -> recruiter -> screening -> assessment -> interview -> offer/closed
- ☐ Ingest both Gmail accounts through redaction boundary
- ☐ Deterministic extraction first; model only for ambiguity
- ☐ Entity resolution for duplicate companies, roles and threads
- ☐ Universal timeline projection
- ☐ Next action + overdue follow-up logic
- ☐ Interview/calendar matching
- ☐ Morning brief + Discord event summaries
- ☐ Golden test set and manual correction workflow
- ☐ No email send or calendar write in v1
- ☐ Dashboard-ready API payloads

Exit: every active application has a stage, evidence trail, confidence and a clear next action.

19 / GOVERNANCE

Project tracking blueprint

Keep architecture alive as decisions, tests and measurable system health.

EVENT HEALTH

ingested / failed / retried / replayed

ATTENTION QUALITY

noticed too early / late / correctly

TRUTH QUALITY

confidence, conflicts, correction rate

AUTONOMY SAFETY

approval rate, denied actions, reversals

MEMORY QUALITY

retrieval usefulness, staleness, deletion

APP VALUE

time saved, actions completed, missed deadlines

WEEKLY OPERATING RHYTHM

Review timeline anomalies / approve schema changes / inspect privacy-zone access / test one failure path / close one roadmap gate / record decisions.

North-star check: Can LILITH explain what she knows, why it matters, what she did and what needs you?



BUILD THE CORE.

THEN GIVE IT A CAREER.

Immediate sequence: OS Core v0.1 -> Career CRM v1 -> Visual Shell. Preserve privacy, visible agency and model independence at every step.

LILITH OS / ARCHITECTURE v0.1

A living blueprint for a personal intelligence operating system